SDG 14: LIFE BELOW WATER PROGRESS REPORT

Sustainable Development Goal 14 (SDG 14), Life Below Water, aims to conserve and sustainably use oceans, seas, and marine resources. It emphasizes preventing and reducing marine pollution, protecting and restoring coastal and marine ecosystems, and promoting scientific knowledge and cooperation to maintain healthy aquatic environments. For higher education institutions, contributing to SDG 14 involves environmental education, coastal clean-up activities, habitat restoration, community-based research, and partnerships that connect academic learning with practical marine conservation.

The City College of Cagayan de Oro (CCCDO) contributed to SDG 14 in 2024 by conducting and participating in mangrove planting and coastal clean-up activities, alongside research examining community vulnerabilities related to flooding and waste management in Barangay Agusan. In 2025, the College continued these efforts through the Baybayani coastal clean-up and mangrove rehabilitation initiatives while expanding its participation in scientific discussions on the blue economy and international sustainability engagements. Taken together, the two reporting periods demonstrate continuity in local marine conservation and a broader range of documented environmental and sustainability activities in 2025.

This report uses CCCDO's 2024 SDG 14 progress report as the baseline and the supplied 2025 narrative and official College articles as the basis for assessing subsequent developments. It examines the activities conducted, their contributions to SDG 14, and their significance for marine ecosystems, coastal communities, student learning, and institutional sustainability. It also distinguishes documented activity outputs from longer-term environmental outcomes to provide a balanced assessment of the College's progress.

Executive Assessment

The central finding of the 2024–2025 comparison is that CCCDO sustained its commitment to direct environmental conservation while broadening its documented marine-related engagement in 2025. The 2024 baseline identified mangrove planting at the Bonbon Mangrove Reserve, coastal clean-up activities along areas including Gusa Coastal Road, and research presented at the 11th International Conference on Teacher Education (ICTED) examining community vulnerabilities in Barangay Agusan. The research connected flooding and waste management challenges with environmental pressures that may affect coastal and marine ecosystems.

In 2025, the College continued its conservation efforts through several Baybayani activities, with measurable outputs documented in two initiatives. Baybayani 2025 collected 30 sacks of garbage across three coastal areas, while Baybayani 2.0 mobilized 60 volunteers to plant 300 mangrove seedlings. Baybayani 3.0, conducted in observance of Earth Day, brought together students, personnel, environmental organizations, government agencies, and community partners for coastal clean-up and environmental awareness activities. These initiatives

demonstrate how collective action can support waste reduction, promote environmental responsibility, and contribute to coastal habitat restoration.

Beyond local conservation, CCCDO participated in the PhilAAST convention and Mindanao Forum on the Blue Economy and engaged in the Global Sustainable Development Congress and the 2025 Times Higher Education Impact Rankings. These activities broadened the College's documented sustainability engagement by connecting local environmental initiatives with scientific discussions, sustainable development priorities, and wider institutional efforts to address global challenges.

Overall, the evidence indicates progress through continuity and diversification. Mangrove planting and coastal clean-up activities remained important components of CCCDO's environmental initiatives, while the 2025 record provided more specific output figures and documented broader participation in scientific and international sustainability activities. These developments matter because marine and coastal ecosystems support biodiversity, fisheries, food security, livelihoods, coastal protection, and community resilience. Protecting these environments also requires cooperation among educational institutions, government agencies, environmental organizations, and local communities.

However, the reported activities and immediate outputs do not, by themselves, establish the extent of long-term environmental improvement. The available records do not quantify overall changes in marine water quality, total marine litter reduction, or mangrove seedling survival over time. Continued monitoring and documentation of environmental outcomes would help CCCDO demonstrate more clearly the lasting contributions of its SDG 14 initiatives.

Key Findings

- The 2024 baseline combined direct environmental action—mangrove planting and coastal clean-up—with research on community vulnerabilities involving flooding and waste management.
- The 2025 record shows continuity through multiple Baybayani initiatives, including coastal clean-up, mangrove rehabilitation, and community-based environmental participation.
- Two 2025 activities provide clear output figures: 30 sacks of garbage collected at three coastal areas, and 300 mangrove seedlings planted by 60 volunteers.
- The 2025 portfolio broadened beyond direct conservation to include scientific dialogue on the blue economy and global sustainability engagement.
- The most defensible interpretation is expanded and more visibly documented activity, not a quantified percentage increase in ocean health or marine biodiversity.

Institutional Context and SDG 14

The connection between CCCDO’s work and SDG 14 is both direct and cross-cutting. Coastal clean-ups directly address litter and debris that can reach marine environments. Mangrove planting supports the restoration of coastal habitat, which is important to marine biodiversity and shoreline stability when seedlings survive and ecosystems are maintained. Community research on flooding and waste management brings a place-based perspective to the land-to-sea pathways through which waste and runoff can affect coastal water. Scientific forums and institutional partnerships provide opportunities to share knowledge, develop

research collaborations, and connect conservation activities to broader discussions of marine resource management.

The report therefore considers three linked pathways: practical conservation activity, community learning and research, and scientific or institutional collaboration. These are different forms of contribution and are not combined into a single impact score. The figures below indicate selected documented activities and outputs; their interpretations explain relevance without overstating unmeasured ecological effects.

2024 Baseline: Conservation and Community-Driven Research

Mangrove planting and coastal habitat protection

The 2024 SDG 14 report identifies CCCDO's participation in a Mangrove Planting Activity at the Bonbon Mangrove Reserve in Cagayan de Oro. The activity involved selected personnel, faculty, and non-teaching staff working alongside local government offices. Mangrove habitat is relevant to SDG 14 because it provides shelter and nursery areas for marine organisms and helps stabilize coastal environments. Mangroves can also function as natural barriers against storm surges and contribute to carbon storage, connecting marine protection with climate resilience. [1][2]

The baseline report also records coastal clean-up activity, including an activity at Gusa Coastal Road. The report describes these efforts as mobilizing the college community to remove marine debris and raise awareness of shoreline protection. The supplied 2024 baseline does not provide a comparable waste-collection quantity, so this activity is best presented as documented participation and pollution-prevention work rather than as a quantified reduction in marine litter. [1]

Research on coastal community vulnerabilities

CCCDO's 2024 research contribution extended the baseline beyond clean-up and planting. At the 11th International Conference on Teacher Education (ICTED), the College presented "Semiotic and Ethnographic Pictorial Analyses on Mapping Community Vulnerabilities: Educational Pathways for Local Disaster Preparedness." The participatory study focused on Barangay Agusan and used photovoice and visual analysis to identify vulnerabilities related to flooding and waste management.

This research matters to SDG 14 because marine pollution often has land-based pathways. Flooding can move waste and debris toward drainage channels and coastal waters, while inadequate waste management can allow plastics and other litter to accumulate in marine habitats. By identifying local vulnerabilities, the study connected academic inquiry with community conditions that shape the health of coastal environments. The baseline therefore combined direct ecological action with an evidence-informed understanding of environmental risks.

2024 baseline strand	Documented activity	Contribution to SDG 14
Coastal habitat restoration	Mangrove planting at the Bonbon Mangrove Reserve; personnel, faculty, and non-teaching staff participated.	Supports mangrove and coastal habitat protection relevant to marine biodiversity.

2024 baseline strand	Documented activity	Contribution to SDG 14
Marine litter reduction	Coastal clean-up activity, including Gusa Coastal Road, recorded in the 2024 report.	Demonstrates direct community involvement in removing marine debris and promoting shoreline stewardship.
Community-driven research	ICTED presentation on mapping community vulnerabilities in Barangay Agusan, including flooding and waste-management concerns.	Connects local evidence and education with understanding of land-based risks to coastal and marine environments.

2025 Progress: Repeated Conservation, Measurable Outputs, and Wider Engagement

Baybayani 2025: Klarex nga Kalimpyo

On February 28, 2025, CCCDO took part in Baybayani 2025: Klarex nga Kalimpyo in Barangay Agusan and Barangay Consolacion. The official report states that nearly 50 faculty and staff members participated alongside local residents. Volunteers covered three coastal areas and collected 30 sacks of garbage. This gives the 2025 record an immediate, countable clean-up output that is not provided for the 2024 baseline activity.

The figure of 30 sacks should be interpreted as a documented output from one activity, not as the total quantity of marine waste removed by CCCDO during the year. Its significance is practical: removing accumulated debris can improve the cleanliness of the covered locations and reduce the amount of litter available to be carried into coastal waters. The participation of staff and residents also illustrates that coastal stewardship is a shared responsibility rather than an activity limited to campus operations.

Baybayani 2.0: Mangrove rehabilitation

Baybayani 2.0 was conducted on March 15, 2025, at Barangay Pangayawan, Gitagum, Misamis Oriental, in collaboration with CLIMBS Life and General Insurance Cooperative and Kusgan Volunteers Inc. The official article reports 60 volunteers from CCCDO and CLIMBS and 300 mangrove seedlings planted.

This activity continues the habitat-restoration strand established in 2024. The 300 seedlings are a clear implementation output and the involvement of 60 volunteers indicates organized participation. Mangrove ecosystems can support marine nurseries, biodiversity, and shoreline stability; these benefits depend on the seedlings surviving and becoming established. The documented planting therefore demonstrates action toward restoration, rather than a verified measure of habitat recovery or marine population change.

Baybayani 3.0: Earth Day coastal clean-up

On April 22, 2025, CCCDO participated in Baybayani 3.0 at the Agusan coastal areas in celebration of Earth Day. The initiative brought together faculty, staff, NSTP student volunteers, environmental organizations, government offices, and community partners, including DENR Region X, CLENRO, Plastik Republic, Opol Community College, KUSGAN

Volunteers Inc., BFP, Coop NATCCO, and the Agusan Barangay Council. The article emphasizes coastal clean-up, environmental awareness, and collaborative stewardship.

Baybayani 3.0 adds continuity to the 2025 clean-up sequence. Together with Baybayani 2025, it indicates that coastal clean-up was not limited to a single event. The published account does not give a separate waste quantity for Baybayani 3.0, so the 30-sack figure from February must not be attributed to the April activity. The value of Baybayani 3.0 lies in collective action, public awareness, and partner coordination around marine and coastal protection.

Scientific engagement and the blue economy

In July 2025, CCCDO participated in the 74th Annual Convention and National Convention of the Philippine Association for the Advancement of Science and Technology (PhilAAST), which included the Mindanao Forum on Blue Economy and Food Security, Nutrition, and Industry Outlook: Moving Forward to Pagtanaw 2050. The forum brought scientists, policymakers, industry representatives, and educators together to discuss marine resources, sustainable food systems, nutrition, and technology-driven development. [7]

This is a different pathway of contribution from coastal clean-up or mangrove planting. Scientific dialogue can help an institution connect its community activities with wider conversations about sustainable marine-resource use, fisheries, coastal livelihoods, and the blue economy. Participation in a forum does not itself establish a new conservation intervention, but it broadens the knowledge and partnership context in which future marine-related teaching, research, and extension can develop.

Global sustainability engagement

CCCDO was represented at the Global Sustainable Development Congress 2025, held from June 16 to 19 at the Istanbul Congress Center in Türkiye. The College also entered the 2025 Times Higher Education Impact Rankings in the 1001–1500 band for the first time. These are institution-wide sustainability milestones rather than direct marine-conservation outputs.

Their relevance to SDG 14 is indirect but meaningful: global sustainability engagement creates opportunities for learning, institutional comparison, and collaboration around the United Nations Sustainable Development Goals. This broader engagement complements local activities by placing coastal conservation within CCCDO's overall sustainability agenda. The direct evidence of marine action remains the clean-up, planting, and research activities, while congress participation and rankings reflect the College's wider institutional orientation toward the SDGs.

2025 Activity Timeline and Documented Outputs

Date	Initiative	Documented output / participation	SDG 14 significance
February 28, 2025	Baybayani 2025: Klarex nga Kalimpyo	Nearly 50 faculty and staff, alongside residents; 30 sacks collected across three coastal areas.	Provides a quantified clean-up output and reinforces collective coastal stewardship.

Date	Initiative	Documented output / participation	SDG 14 significance
March 15, 2025	Baybayani 2.0: Mangrove Rehabilitation	60 volunteers planted 300 mangrove seedlings.	Continues the 2024 habitat-restoration approach through a measurable planting activity.
April 22, 2025	Baybayani 3.0: Coastal Clean-Up	Faculty, staff, NSTP students, government offices, environmental organizations, and community partners participated.	Sustains clean-up engagement and broadens collaboration around coastal protection.
June 16–19, 2025	Global Sustainable Development Congress	CCCDO represented by its president; first participation in the 2025 THE Impact Rankings, 1001–1500 band.	Places marine conservation within the College's wider institutional sustainability engagement.
July 16–18, 2025	PhilAAST convention and Blue Economy Forum	CCCDO delegation participated in science and technology discussions on the blue economy and food security.	Connects marine conservation with scientific dialogue and sustainable marine-resource use.

Figures and Comparative Analysis

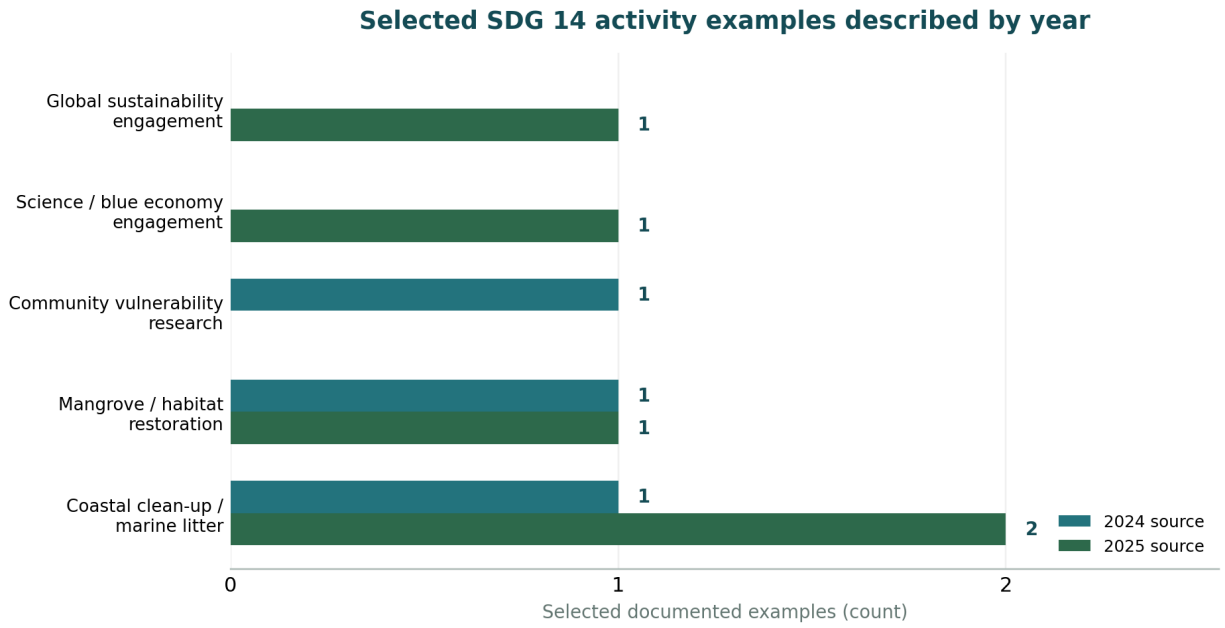


Figure 1. Selected documented SDG 14 activity examples in the 2024 and 2025 source materials.

Interpretation: The chart shows continuity in two direct-action categories: coastal clean-up and mangrove restoration appear in both reporting periods. The 2024 source also specifically describes community vulnerability research presented at ICTED, while the 2025 narrative adds more distinct science and global engagement examples through the PhilAAST Blue Economy Forum and the Global Sustainable Development Congress. The number of examples reflects how the supplied narratives describe activity, not a complete institutional inventory, spending total, or measure of ecological impact.

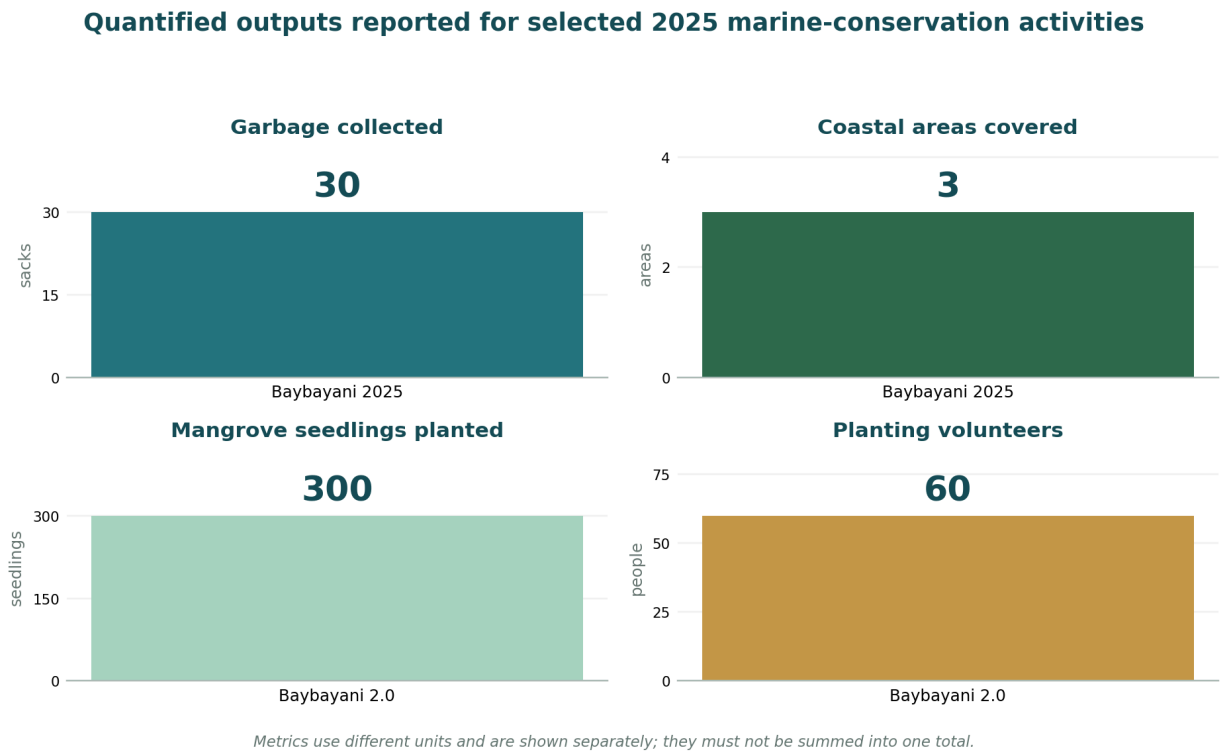


Figure 2. Quantified outputs reported for Baybayani 2025 and Baybayani 2.0. The four measures use different units and are presented separately.

Interpretation: The 2025 reports provide two useful measures of direct action. Baybayani 2025 collected 30 sacks of garbage across three coastal areas, and Baybayani 2.0 planted 300 mangrove seedlings with 60 volunteers. These figures make the immediate scale of selected activities more visible than the 2024 baseline, which did not report equivalent quantities. The measures should not be combined into one numerical total or interpreted as proof of lasting improvements in water quality, biodiversity, or restored habitat.

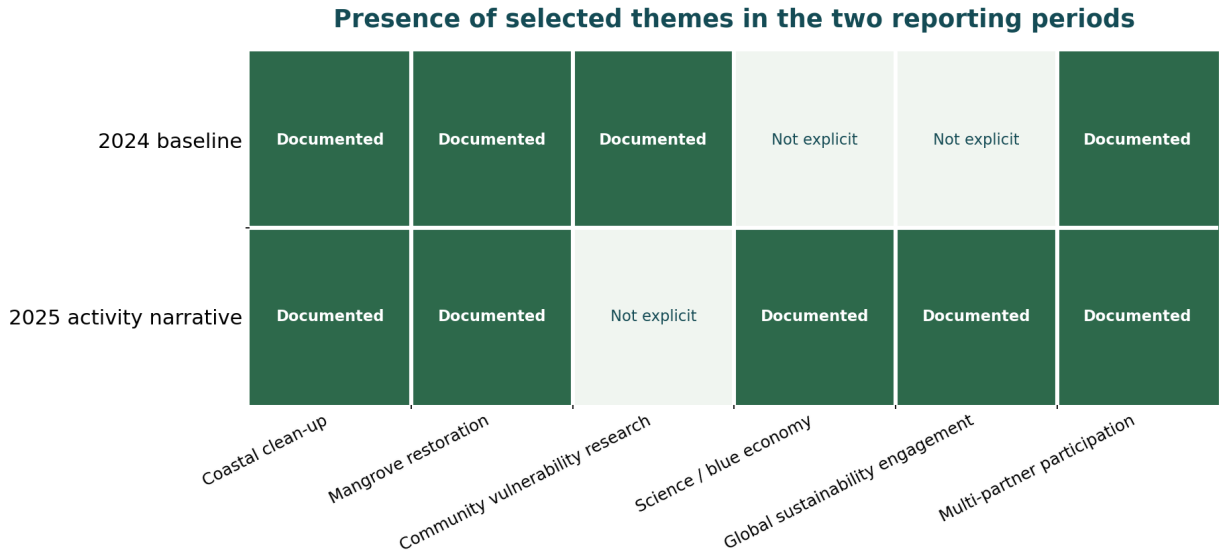


Figure 3. Selected themes explicitly described in the supplied baseline and annual narrative. "Not explicit" does not mean no activity occurred.

Interpretation: Both years document direct coastal action and habitat restoration, reinforcing the idea that marine stewardship was sustained across the reporting period. The 2024 report places greater emphasis on the relationship between local vulnerability research and marine protection. The 2025 narrative shows an expanded record of organized conservation activities and external scientific or institutional engagement. The pattern illustrates broadening in documented scope rather than a quantified increase in ecological results.

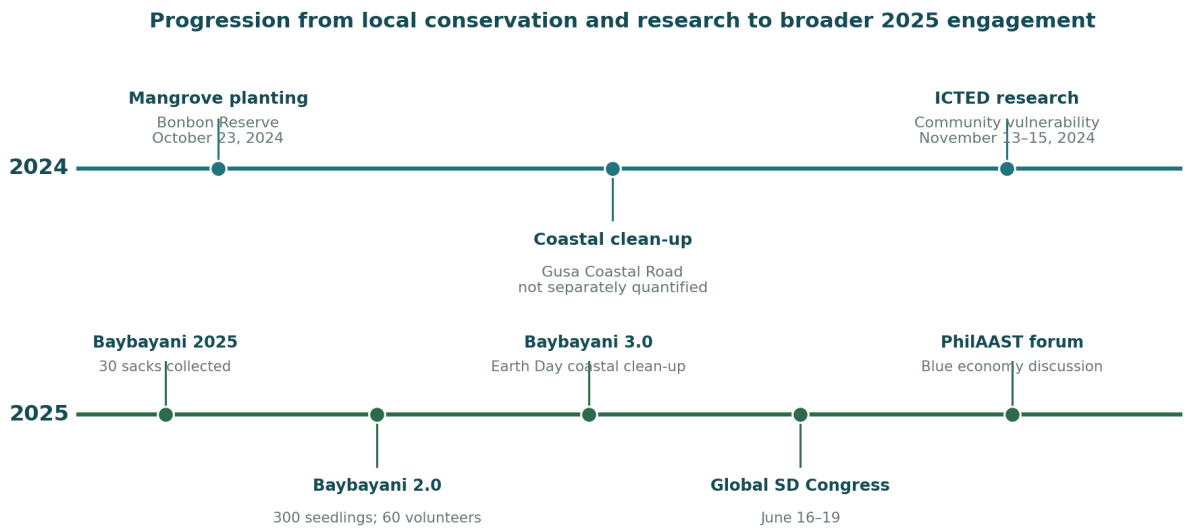


Figure 4. Timeline of selected 2024 and 2025 milestones included in the source documents.

Interpretation: The timeline highlights continuity across years: a 2024 mangrove activity and community research presentation provide the baseline, while the 2025 sequence includes clean-ups in February and April, mangrove planting in March, and wider science and sustainability engagement in June and July. The pattern matters because repeated activities

create more opportunities for student participation, community coordination, and the linking of practical conservation with research and external knowledge networks.

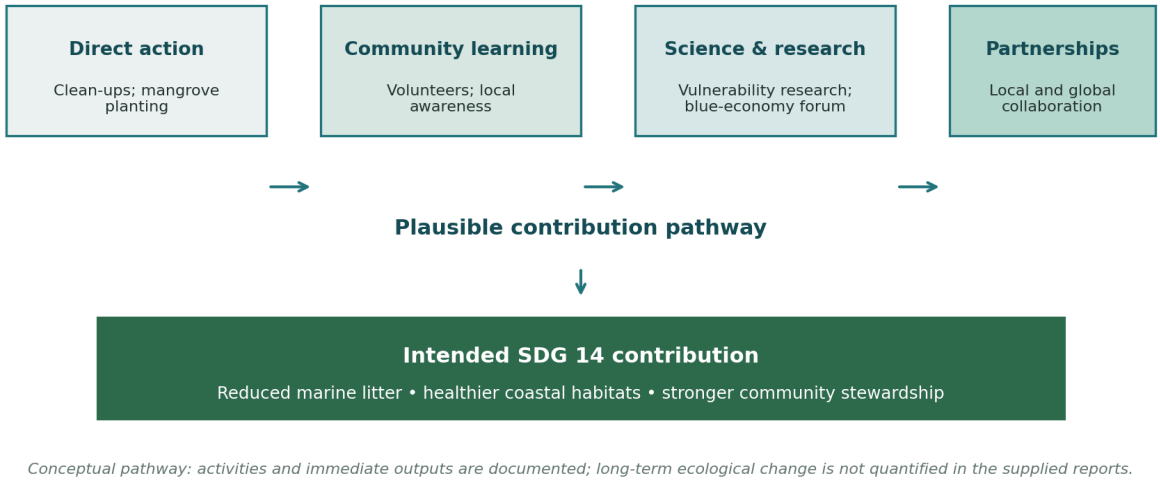


Figure 5. Conceptual pathway connecting the reported activities to their intended SDG 14 contribution.

Interpretation: The diagram explains how direct action, community learning, research, and partnerships may reinforce one another. Clean-ups generate immediate waste-removal outputs; mangrove planting initiates habitat restoration; community research identifies vulnerabilities that can inform local planning; and science forums and global engagement create channels for knowledge exchange. Together, these activities are relevant to pollution reduction, healthier coastal ecosystems, and stronger environmental stewardship. The pathway is conceptual and does not represent measured changes in marine condition.

Impact and Institutional Synergy

The comparison indicates that CCCDO's contribution to SDG 14 brings together practical environmental work and knowledge-based engagement. In 2024, mangrove planting and coastal clean-up demonstrated direct participation in conservation, while the ICTED research presentation connected community vulnerabilities with flooding and waste management. In 2025, repeated Baybayani activities expanded the direct-action strand and yielded clear output counts for waste collection and seedling planting. Participation in the PhilAAST Blue Economy Forum added a scientific and resource-management perspective, while the Global Sustainable Development Congress strengthened the College's broader sustainability engagement.

These strands are complementary. Local action can improve the immediate condition of selected sites and make environmental responsibility visible to students and residents. Research can help explain why certain places are vulnerable and how land-based practices relate to coastal conditions. Scientific networks can connect institutional learning with wider debates on sustainable fisheries, marine resources, and blue-economy development. Partnerships help bring together the participants, technical knowledge, and organizational capacity required for coordinated activity.

Dimension	2024 evidence	2025 evidence	Comparative interpretation
Direct marine conservation	Mangrove planting at Bonbon and coastal clean-up activity identified in the baseline report.	Baybayani 2025 and Baybayani 3.0 coastal clean-ups; Baybayani 2.0 mangrove planting.	The direct-action strand continued, with clearer quantitative outputs for two selected 2025 activities.
Research and knowledge	ICTED presentation on community vulnerabilities involving flooding and waste management.	Participation in the PhilAAST convention and Mindanao Forum on the Blue Economy.	The knowledge strand shifted from local vulnerability research in the baseline to broader scientific dialogue on marine resources and the blue economy in 2025.
Community participation	College personnel, faculty, and non-teaching staff participated in conservation activities; research focused on Barangay Agusan.	Faculty, staff, NSTP students, residents, volunteers, and multiple partner organizations took part in Baybayani initiatives.	Participation remained important, with wider partner involvement described in the 2025 coastal clean-up narrative.
Institutional and global engagement	The 2024 report documents public reporting of conservation and research activities.	Global Sustainable Development Congress participation and first entry in THE Impact Rankings.	The 2025 narrative places local marine initiatives within a broader institutional sustainability context.
Quantified outputs	No comparable waste or mangrove counts supplied in the 2024 SDG 14 report.	30 sacks of garbage, three coastal areas, 300 seedlings, and 60 volunteers reported for selected initiatives.	The 2025 source is more specific about selected immediate outputs; these cannot be converted into an annual percentage increase without equivalent 2024 measures.

Alignment with SDG 14 Themes

The initiatives are most directly connected to marine pollution prevention, coastal and marine ecosystem protection, and scientific knowledge and cooperation. This alignment describes the purpose and contribution pathway of the activities; it does not imply that each initiative independently achieved an SDG-level outcome.

SDG 14 theme	CCCDO initiatives	How the contribution matters
Preventing marine pollution	2024 and 2025 coastal clean-ups; Baybayani 2025 reported 30 sacks of garbage collected.	Removes visible litter from selected coastal sites and promotes responsible waste handling and shared stewardship.
Protecting coastal and marine ecosystems	Mangrove planting in Bonbon in 2024; Baybayani 2.0 planting of 300 seedlings in 2025.	Supports the restoration of habitat that can provide nursery areas for marine organisms and help stabilize shorelines.
Scientific knowledge and awareness	ICTED community vulnerability research; PhilAAST Blue Economy Forum.	Connects academic learning with community risks and discussions on sustainable marine-resource use.
Partnerships and community participation	Coastal clean-ups with residents, NSTP students, local agencies, environmental organizations, and volunteers.	Combines institutional, community, and technical contributions around coastal conservation.
Cross-cutting sustainability	Global Sustainable Development Congress and THE Impact Rankings participation.	Places local marine conservation within CCCDO's broader institutional engagement with the Sustainable Development Goals.

Why CCCDO's Progress Matters

The progress matters first because coastal and marine ecosystems are important to biodiversity and the natural systems that support coastal communities. Mangroves provide habitat for marine organisms and can help stabilize shorelines. Clean-ups address visible waste that can threaten wildlife and degrade coastal environments. The documented 2025 figures provide concrete examples of CCCDO's participation: 30 sacks of garbage removed across three coastal areas and 300 mangrove seedlings planted by 60 volunteers.

Second, marine conservation supports people as well as ecosystems. Healthy coastal environments underpin fisheries, food security, livelihoods, recreation, and local identity. Scientific discussion of the blue economy helps connect environmental protection with the sustainable use of marine resources. Research into flooding and waste-management vulnerabilities also demonstrates how community conditions on land can influence coastal pollution and resilience.

Third, the activities strengthen the educational role of the College. Students and personnel participate in hands-on environmental work, while community research and science forums connect local experiences with broader knowledge. Partnerships with community organizations and environmental groups help make conservation a collective activity. This is

particularly relevant for higher education, where the benefits of teaching and research extend beyond classroom instruction into civic responsibility and community engagement.

Overall, CCCDO's documented progress from 2024 to 2025 is characterized by continuing direct conservation work, more specific reported outputs for selected 2025 activities, and an expanded connection to scientific and global sustainability networks. The evidence supports a conclusion of sustained and diversified engagement with SDG 14; it does not establish a quantified change in overall ocean health or long-term ecosystem condition.

Conclusion

The 2024–2025 comparison shows that CCCDO maintained a practical commitment to Life Below Water while broadening its documented engagement. The 2024 baseline combined mangrove planting, coastal clean-up, and community-driven research on flooding and waste management. In 2025, Baybayani 2025 and Baybayani 3.0 continued coastal clean-up work, Baybayani 2.0 advanced mangrove planting, and the College's participation in the PhilAAST Blue Economy Forum and the Global Sustainable Development Congress connected local conservation with wider scientific and institutional sustainability discussions.

The two quantified 2025 outputs—30 sacks of garbage collected across three coastal areas and 300 mangrove seedlings planted by 60 volunteers—show tangible implementation of selected activities. Their significance lies in demonstrating practical environmental action and organized community participation. Taken together, the reports portray a higher education institution contributing to marine conservation through action, research, education, and collaboration, in support of healthier coastal environments and more environmentally informed communities.

Sources and Verification Notes

[1] City College of Cagayan de Oro. (n.d.). Progress Report on SDG 14: Life Below Water [Institutional report]. https://citycollegecdo.edu.ph/dist/SDG/SDG14-Life-Below-Water-Progress-report.pdf

[2] City College of Cagayan de Oro. (2024, October 23). City College of Cagayan de Oro Participates on Mangrove Planting Activity 2024. https://citycollegecdo.edu.ph/article/1184

[3] City College of Cagayan de Oro. (2024, November 18). City College of CDO Presents Community-Driven Disaster Resilience Study at ICTED 2024. https://citycollegecdo.edu.ph/article/803

[4] City College of Cagayan de Oro. (2025, February 28). Baybayani 2025: Klarex nga Kalimpyo. https://citycollegecdo.edu.ph/article/1150

[5] City College of Cagayan de Oro. (2025, March 20). Baybayani 2.0: A Mangrove Rehabilitation Initiative. https://citycollegecdo.edu.ph/article/1149

[6] City College of Cagayan de Oro. (2025, April 22). Baybayani 3.0: A Coastal Clean-Up Initiative. https://citycollegecdo.edu.ph/article/1148

[7] City College of Cagayan de Oro. (2025, July 18). CCCDO Joins 74th PhilAAST Annual Convention and Mindanao Forum on Blue Economy and Food Security. https://citycollegecdo.edu.ph/article/1104

[8] City College of Cagayan de Oro. (2025, June 18). City College of CDO Steps onto the Global Stage for Sustainability and Impact. https://citycollegecdo.edu.ph/article/1052

[9] City College of Cagayan de Oro. (2023, September 21). City College of Cagayan de Oro Participates in Coastal Cleanup at Gusa Coastal Road. https://citycollegecdo.edu.ph/article/836

Source alignment note: The supplied 2024 SDG 14 PDF gives September 19, 2024, for the Bonbon mangrove activity in its narrative and table, while the official article is dated October 23, 2024. This report uses the official article date for the timeline. The baseline PDF describes coastal clean-up activity at Gusa Coastal Road; the separately listed Gusa article is dated 2023, so it is cited as historical context rather than treated as a quantified 2024 output.